REPORT OF NOISE MONITORING
ENVIRONMENT MONITORING FOR KU CAMPUS
ABU DHABI, UAE

Request No.	**EQ21003369**
Report No.	**ER22000026**
Revision No.	0
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PREPARED FOR:
Khalifa University for Science and Technology

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Request No: EQ21003369
Report No: ER22000026
Report Date: 13/01/2022

SUBJECT	:	NOISE MONITORING REPORT
PROJECT	:	Environment Monitoring for KU Campus
PROJECT NO	:	Not Applicable
OWNER	:	Khalifa University
CONSULTANT	:	Not Applicable
LOCATION	:	Khalifa University Main Campus, Abu Dhabi
TESTING DATES	:	19-21/12/2021

1.0 INTRODUCTION

Pursuant to your request, ref. RFQ 6391231 dated 12th July 2021 related to the above subject, Arab Center for Engineering Studies is pleased to submit the obtained results of noise level monitoring carried out on the above project on 19-21/12/2021.

2.0 PURPOSE

The purpose of the study is to measure the Noise level during a specific monitoring period and assess its compliance with requirements of the Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF), Code of Practice, Cop3.0, Occupational Noise, Version 3.0, App.2

3.0 SCOPE OF WORKS

The scope of work consists of the following:

- Conducting noise level measurement by taking sound level meter readings on minutes basis for monitoring period and location(s) specified by Client (see Cl.4 and Cl.5 for the locations and monitoring details).
- Estimate of the "day time" averages Occupational Noise Exposure Limits and comparing with requirements of the Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF), Code of Practice, Cop3.0, Occupational Noise, Version 3.0, App.2.

4.0 LOCATION(S) AND SAMPLING DETAILS

Location Description	Visible Noise Sources Present	Sampling Height
Building G (G01015)	-	1.5m above the ground level
Building G (G03008)	-	1.5m above the ground level
Building S (S00001)	-	1.5m above the ground level
Building R Chemistry Laboratory (R02031)	Fume Hood, Lab Activities	1.5m above the ground level
Building R Chemistry Faculty Offices (R02035)	-	1.5m above the ground level
Building R (R01007)	-	1.5m above the ground level
Building M First floor	-	1.5m above the ground level
Building L (L01007)	-	1.5m above the ground level
Building L (L01016)	Orbital Incubator, Fume Hood, Fridge, Routine Activities	1.5m above the ground level
Building E (Library)	-	1.5m above the ground level
Building E (4th Floor)	-	1.5m above the ground level
Building E (E04006A)	-	1.5m above the ground level
Building C Aircraft Design Lab C01014	-	1.5m above the ground level
Building C (C01028)	-	1.5m above the ground level
Building C (C00026)	Milling Machine, Mechanical Activities On-going	1.5m above the ground level
Building D Nurse room (D00009)	-	1.5m above the ground level
Building B (B01009)	-	1.5m above the ground level
Building B Dry Laboratory-2 (B00118)	-	1.5m above the ground level
Building A Dry Laboratory-1 (A00125)	Air Conditioner Sounds	1.5m above the ground level
Building E Director in Institutional Effectiveness Office (E04001)	-	1.5m above the ground level

**Photo of monitoring location(s) are presented in Appendix A.*

5.0 MONITORING DETAILS

Location Description	Parameter	Sampling Dates and Time	Data Capture Rate%
Building G (G01015)	**Background Noise Level L_{90A}**	19/12/2021 11:35 - 11:45	100%
	Source Noise L_{Aeq}	19/12/2021 11:45 - 12:00	100%
Building G (G03008)	**Background Noise Level L_{90A}**	19/12/2021 12:35 -13:45	100%
	Source Noise L_{Aeq}	19/12/2021 12:45 - 13:00	100%
Building S (S00001)	**Background Noise Level L_{90A}**	19/12/2021 13:35 -13:45	100%
	Source Noise L_{Aeq}	19/12/2021 13:45 - 14:00	100%
Building R Chemistry Laboratory (R02031)	**Background Noise Level L_{90A}**	20/12/2021 13:00 - 13:09	100%
	Source Noise L_{Aeq}	20/12/2021 13:10 - 13:24	100%
Building R Chemistry Faculty Offices (R02035)	**Background Noise Level L_{90A}**	20/12/2021 11:55 - 12:05	100%
	Source Noise L_{Aeq}	20/12/2021 12:05 - 12:20	100%
Building R (R01007)	**Background Noise Level L_{90A}**	20/12/2021 13:28 - 13:37	100%
	Source Noise L_{Aeq}	20/12/2021 13:38 - 13:52	100%
Building M First floor	**Background Noise Level L_{90A}**	20/12/2021 12:25 - 12:35	100%
	Source Noise L_{Aeq}	20/12/2021 12:35 - 12:50	100%
Building L (L01007)	**Background Noise Level L_{90A}**	20/12/2021 14:04 - 14:13	100%
	Source Noise L_{Aeq}	20/12/2021 14:14 - 14:28	100%
Building L (L01016)	**Background Noise Level L_{90A}**	20/12/2021 14:04 -14:13	100%
	Source Noise L_{Aeq}	20/12/2021 14:14 - 14:28	100%
Building E (Library)	**Background Noise Level L_{90A}**	20/12/2021 13:00 - 13:10	100%
	Source Noise L_{Aeq}	20/12/2021 13:10 - 13:25	100%
Building E (4th Floor)	**Background Noise Level L_{90A}**	20/12/2021 13:40 - 13:50	100%
	Source Noise L_{Aeq}	20/12/2021 13:50 - 14:05	100%
Building E (E04006A)	**Background Noise Level L_{90A}**	20/12/2021 15:28 - 15:37	100%
	Source Noise L_{Aeq}	20/12/2021 15:38 - 15:52	100%

Building C Aircraft Design Lab C01014	**Background Noise Level L_{90A}**	20/12/2021 14:15 - 14:25	100%
	Source Noise L_{Aeq}	20/12/2021 14:25 - 14:40	100%
Building C (C01028)	**Background Noise Level L_{90A}**	21/12/2021 09:47 - 09:56	100%
	Source Noise L_{Aeq}	21/12/2021 09:57 - 10:11	100%
Building C (C00026)	**Background Noise Level L_{90A}**	21/12/2021 08:45 - 08:55	100%
	Source Noise L_{Aeq}	21/12/2021 08:55 - 09:10	100%
Building D Nurse room (D00009)	**Background Noise Level L_{90A}**	21/12/2021 10:17 - 10:26	100%
	Source Noise L_{Aeq}	21/12/2021 10:27 - 10:41	100%
Building B (B01009)	**Background Noise Level L_{90A}**	21/12/2021 09:20 - 09:30	100%
	Source Noise L_{Aeq}	21/12/2021 09:30 - 09:45	100%
Building B Dry Laboratory-2 (B00118)	**Background Noise Level L_{90A}**	21/12/2021 10:58 - 11:07	100%
	Source Noise L_{Aeq}	21/12/2021 11:08 - 11:22	100%
Building A Dry Laboratory-1 (A00125)	**Background Noise Level L_{90A}**	21/12/2021 09:51 - 10:00	100%
	Source Noise L_{Aeq}	21/12/2021 10:00 - 10:15	100%
Building E Director in Institutional Effectiveness Office (E04001)	**Background Noise Level L_{90A}**	21/12/2021 11:35 - 11:44	100%
	Source Noise L_{Aeq}	21/12/2021 11:45 - 11:59	100%

6.0 MONITORING METHOD

Sound Level measurements were made with Sinus Tango Precision Sound Level Meter, Class 1 in accordance with ACES SOP047 based on ISO1996. The sound level meter was set at fast response in A-weighted frequency mode over the specified measurement period of 15 minutes. A one-half-inch microphone was mounted on a tripod approximately 1.5m above ground level with microphone pointed at the proper angle relative to an imaginary line to the source. Care was exercised to assure the measurement position was free from excess reflections from walls, columns, or other obstructing objects that may cause interference, and as much as possible, from heavy winds and other adverse conditions.

The analyzer and calibrator have valid annual calibration performed by ISO17025 accredited laboratory. On site calibrations were also performed before and after each measurement series. The calibration was done at 94dB and the calibration level variation was below ± 1.0 dB. (In the event that the deviation from the first calibration is greater than 1 dB (A) then the results obtained will not be valid and the assessment will have to be repeated).

Background noise assessment LA90 was made as much as possible in the absence of source noise. The assessment on the noise from the source under concern was completed by measurement of equivalent A-weighted sound level (LAeq) at each measurement location for a period of 15 minutes.

7.0 SUMMARY OF MONITORING RESULTS

NOISE RESULTS

Start / End Time: 19/12/2021 11:45 - 19/12/2021 12:00

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building G (G01015)	-	58 (Background Noise)	-	-	-	60	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 19/12/2021 12:45 - 19/12/2021 13:00

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building G (G03008)	-	38 (Background Noise)	-	-	-	38	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 19/12/2021 13:45 - 19/12/2021 14:00

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building S (S00001)	-	40 (Background Noise)	-	-	-	48	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 13:10 - 20/12/2021 13:24

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building R Chemistry Laboratory (R02031)	-	51 (Background Noise)	-	-	-	52	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 12:05 - 20/12/2021 12:20

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building R Chemistry Faculty Offices (R02035)	-	42 (Background Noise)	-	-	-	45	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 13:38 - 20/12/2021 13:52

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building R (R01007)	-	40 (Background Noise)	-	-	-	41	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 12:35 - 20/12/2021 12:50

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building M First floor	-	39 (Background Noise)	-	-	-	44	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 14:14 - 20/12/2021 14:28

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building L (L01007)	-	34 (Background Noise)	-	-	-	36	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 13:10 - 20/12/2121 13:25

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building L (L01016)	-	62 (Background Noise)	-	-	-	63	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 15:02 - 20/12/2021 15:16

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building E (Library)	-	53 (Background Noise)	-	-	-	54	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 13:50 - 20/12/2021 14:05

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building E (4th Floor)	-	39 (Background Noise)	-	-	-	39	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 15:38 - 20/12/2021 15:52

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building E (E04006A)	-	33 (Background Noise)	-	-	-	34	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 20/12/2021 14:25 - 20/12/2021 14:40

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building C Aircraft Design Lab C01014	-	38 (Background Noise)	-	-	-	40	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 21/12/2021 09:57 - 21/12/2021 10:11

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building C (C01028)	-	43 (Background Noise)	-	-	-	45	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 21/12/2021 08:55 - 21/12/2021 09:10

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building C (C00026)	-	54 (Background Noise)	-	-	-	68	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 21/12/2021 10:27 - 21/12/2021 10:41

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building D Nurse room (D00009)	-	32 (Background Noise)	-	-	-	38	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 21/12/2021 09:30 - 21/12/2021 09:45

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building B (B01009)	-	34 (Background Noise)	-	-	-	39	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 21/12/2021 11:08 - 21/12/2021 11:22

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building B Dry Laboratory-2 (B00118)	-	49 (Background Noise)	-	-	-	49	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 21/12/2021 10:00 - 21/12/2021 10:15

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building A Dry Laboratory-1 (A00125)	-	59 (Background Noise)	-	-	-	60	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

Start / End Time: 21/12/2021 11:45 - 21/12/2021 11:59

Location	LA_{10}	LA_{90}	LAE	LAF_{max}	LAF_{min}	L(A)eq	LC_{peak}	LAS	LAS_{max}	LAS_{min}
	dB (A)									
Building E Director in Institutional Effectiveness Office (E04001)	-	37 (Background Noise)	-	-	-	38	-	-	-	-
*Noise Limits db(A), See [Ref.]**	-	-	-	-	-	*100 Max. allowable*		-	-	-

**[Ref]: Abu Dhabi Occupational Safety and Health System Framework (OSHAD-SF), Code of Practice, Cop3.0, Occupational Noise, Version 3.0, App.2*

METEOROLOGICAL DATA

Location Id No.	Ambient Temperature	Relative Humidity
	(Min./Max.)ºC	%
Building G (G01015)	20.7 - 24.3	46
Building G (G03008)	19.9 - 22.1	41
Building S (S00001)	20.7 - 25.1	43
Building R Chemistry Laboratory (R02031)	22.7 - 24.1	45
Building R Chemistry Faculty Offices (R02035)	22.5 - 23.8	43
Building R (R01007)	20.9 - 23.6	44
Building M First floor	19.5 - 24.1	42
Building L (L01007)	20.5 - 24.1	42
Building L (L01016)	21.1 - 24.0	46
Building E (Library)	21.9 - 24.7	47
Building E (4th Floor)	21.7 - 23.9	45
Building E (E04006A)	19.4 - 22.6	40
Building C Aircraft Design Lab C01014	19.7 - 23.4	42
Building C (C01028)	21.0 - 23.9	43

Building C (C00026)	22.2 - 24.5	41
Building D Nurse room (D00009)	20.5 - 22.8	42
Building B (B01009)	19.2 - 22.1	40
Building B Dry Laboratory-2 (B00118)	19.9 - 23.8	45
Building A Dry Laboratory-1 (A00125)	20.5 - 23.7	45
Building E Director in Institutional Effectiveness Office (E04001)	20.7 - 23.5	45

*Ref: www.ncms.ae

8.0 IMPORTANT NOTES

- The results presented in this report relate only to the specific location and specific monitoring period. It should be noted that they may be subject to variation caused by variation of activities at/or nearby the site and/or change of environmental conditions.
- This report shall not be reproduced except in full, without written approval of ACES.
- Unless otherwise stated, Client and project details as well as sampling and sample identification information included in this report are provided by the Client. ACES shall not be liable for any incorrect results arising from unclear, erroneous, incomplete, misleading or false information provided by the Client.
- Specification Limits included in this report are provided for information, with no obligation to ACES for providing a statement of conformity.

9.0 APPENDICES

Appendix A - Photograph of Monitoring Location(s)

Appendix B - Project Resources

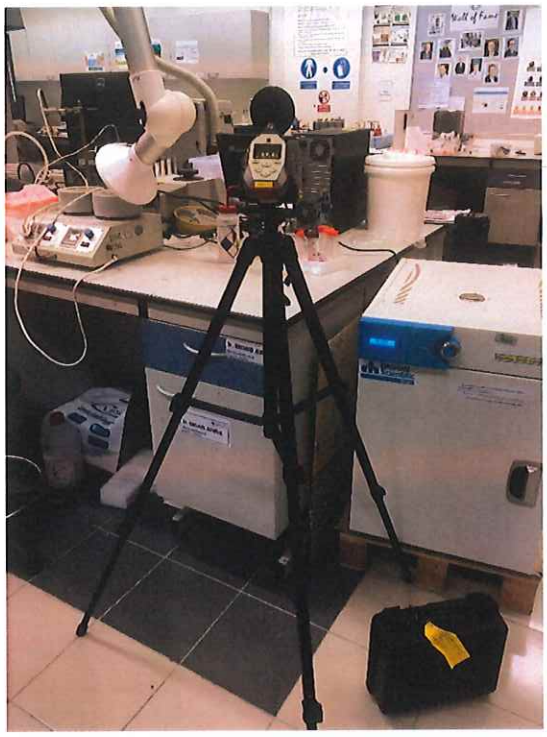
Appendix C - Definitions and Abbreviations Used

Sincerely yours,
ARAB CENTER FOR ENGINEERING STUDIES

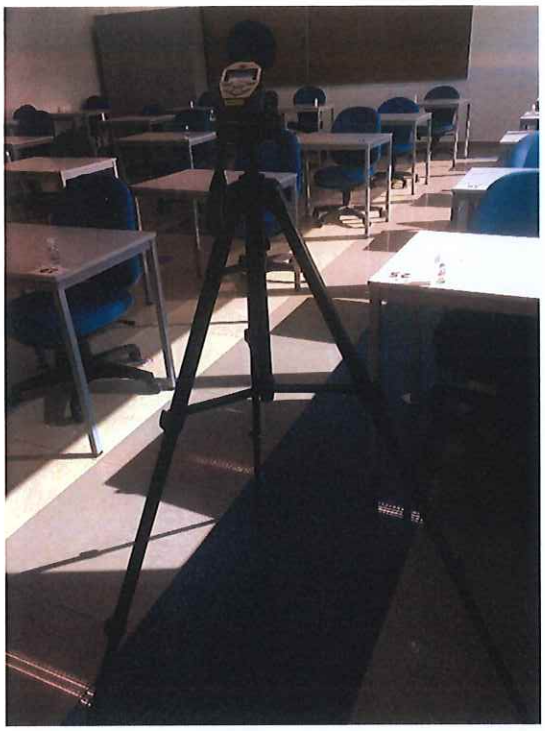
Environmental and Chemical Department Manager

Arab Center For Engineering Studies المركز العربي للدراسات الهندسية

APPENDIX A –PHOTOGRAPH OF MONITORING LOCATION(S)



Building G (G01015)



Building G (G03008)

Building S (S00001)

Building R Chemistry Laboratory (R02031)

Building R Chemistry Faculty Offices (R02035)

Building R (R01007)

Building M First floor

Building L (L01007)

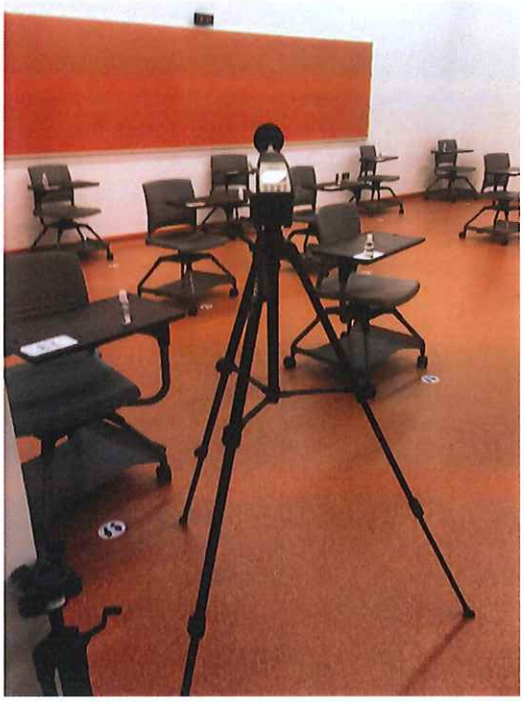
Building L (L01016)

Building E (Library)

Building E (4th Floor)

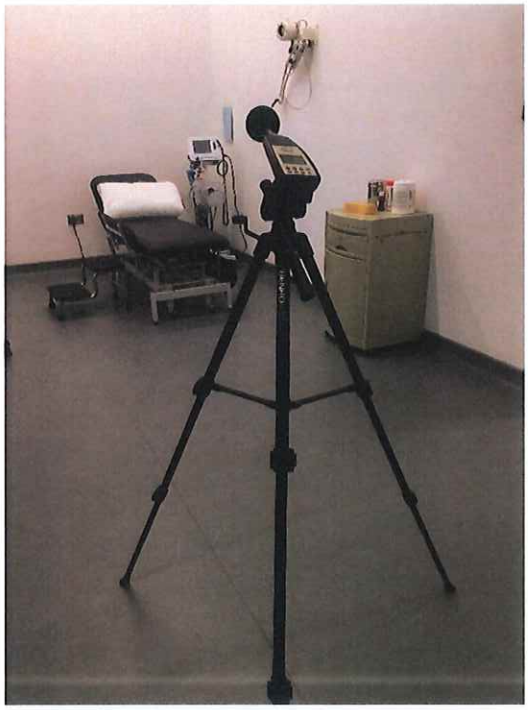
Building E (E04006A)

Building C Aircraft Design Lab C01014

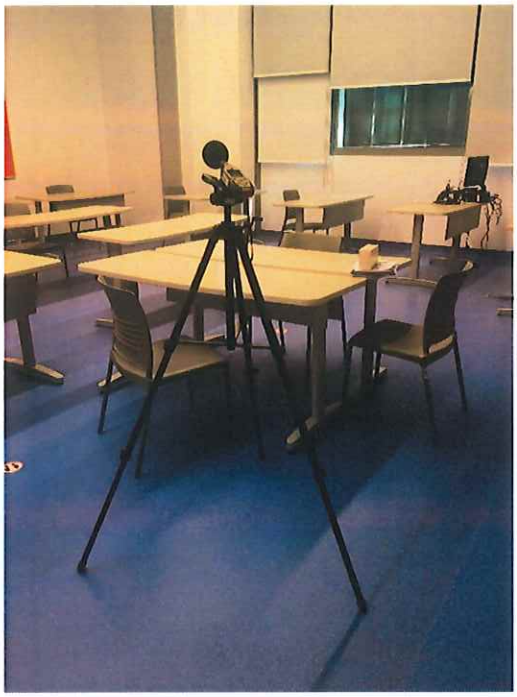


Building C (C01028)

Building C (C00026)



Building D Nurse room (D00009)



Building B (B01009)

Building B Dry Laboratory-2 (B00118)

Building A Dry Laboratory-1 (A00125)

Building E Director in Institutional Effectiveness Office (E04001)

APPENDIX B

PROJECT RESOURCES

Equipment Used

Equipment	Equipment ID
Sound Level Meter	V24/1/4H-A
Calibrator	V24/1/4H-B
RH meter	V16/4/100H
Thermometer	V16/4/100H
Anemometer	V46/1/1H

Calibration Certificates of the above equipments are available up on request.

ACES Monitoring Personnel

Position	Name
Supervisor	Samir Elrifai
Environmental Monitoring Technician	Qudus Olatunde

APPENDIX C

DEFINITIONS AND ABBREVIATIONS USED

A Weighing	A standard weighting of the audible frequencies designed to reflect the response of the human ear
db (A)	Decibels A weighted
LAeq	Equivalent continuous sound pressure level. A measure of the average sound pressure level
LAF_{max}	The minimum sound level with A weighting
LAF_{min}	The maximum sound level with A weighting
LAF	Sound level with 'A' frequency weighting and fast time weighting
LAE	Sound Exposure Level (SEL) with 'A' frequency weighting
LC_{peak}	Peak sound pressure level with 'C' frequency weighting
LAS	Sound level with 'A' frequency weighting and slow time weighting
LAS_{max}	The maximum Sound Level with 'A', frequency weighting and Slow Time weighting
LAS_{min}	The minimum Sound Level with 'A', frequency weighting and Slow Time weighting
LA_{10} & LA_{90}	Background sound pressure level

-END OF REPORT-